



**Ref:** SMTTP/MECH MAINT-/2024-25/LMI\_DA/545202

**Date:** 28/09/2024(18:56:53)

**Subject:** Approval for LMI on 'Deaerator Storage Vessels - Reinspection and Monitoring Requirements'

CC-OS has revised the OGN with ref.no: OGN/OPS/MECH/012.

So, the LMI on 'Deaerator Storage Vessels - Reinspection and Monitoring Requirements' with Ref no: LMI/OGN/OPS/MECH/012/ is due for revision and necessary changes are done in the LMI.

LMI is hereby put up for approval of Competent Authority.

Changes incorporated in this revision (Rev: 7), have been mentioned as below

| Sl No | LMI/OGN/OPS/MECH/012/ ISSUE-1, REV.-06, Jul. 2022 | LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-07, Sep. 2024                      |
|-------|---------------------------------------------------|-----------------------------------------------------------------------|
| 1.    | -                                                 | New Clause 4.1.1 (h): Water spillage through vent have been included. |
| 2.    | -                                                 | New Clause 4.1.1 (i): Civil foundation checking has been included.    |
| 3.    | -                                                 | New Clause 4.1.1 (j): Standpipe inspection have been included.        |
| 4.    | -                                                 | Appendix-2: Checklist has been included.                              |

Enclosures: -

01. LMI on 'Deaerator Storage Vessels - Reinspection and Monitoring Requirements' with Ref no: LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-07, Sep. 2024

02. COS-ISO-00-OGN/OPS/MECH/012 Rev. No.: 3, Aug. 2024

**Submitted By :**



MECH MAINT- TURBINE Department  
SIMHADRI



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**On:**01/10/2024(11:29:25) **Ref:**SMTTP/MECH MAINT-/2024-25/LMI\_DA/545202

Unauthorized Use Prohibited

**Note No:2**

LMI is reviewed in line with revised the OGN with ref.no: OGN/OPS/MECH/012 issued by CC-OS.  
Put up for approval of competent authority.

**Submitted by:**

के पी रमेश - वरिष्ठ प्रबंधक

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**On:**01/10/2024(12:17:21) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:3**

**Submitted by:**

रश्मि रंजन सतपथी - उप महाप्रबंधक

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**On:**04/10/2024(11:09:33) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:4**

As discussed

**Submitted by:**

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**On:**04/10/2024(11:58:01) **Ref:**SMTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:5**

**Submitted by:**

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**On:**04/10/2024(12:38:58) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:6**

May be approved

**Submitted by:**

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**On:**04/10/2024(15:51:44) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:7**

LMI on 'Deaerator Storage Vessels - Reinspection and Monitoring Requirements' is in line wiith the revised OGN (OGN/OPS/MECH/012).

**Submitted by:**

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**On:**04/10/2024(16:05:07) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:8**

Please discuss

**Submitted by:**

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**On:**07/10/2024(19:36:41) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:9**

The LMI was reviewed. It is in line with the revised OGN (OGN/OPS/MECH/012).

May please be approved.

**Submitted by:**

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**On:**09/10/2024(19:56:31) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:10**

May please be approved.

**Submitted by:**

प्रसन्ता कुमार जेना - महाप्रबंधक

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**On:**10/10/2024(15:47:16) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**Note No:11**

**Approved & Submitted by:**

संजय कुमार सिन्हा - मुख्य महाप्रबंधक

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**On:**15/10/2024(18:34:15) **Ref:**SMTTPP/MECH MAINT-/2024-25/LMI\_DA/545202

**NTPC**

**SIMHADRI**

**LOCATION MANAGEMENT INSTRUCTION**

**Deaerator Storage Vessels - Reinspection and Monitoring  
Requirements**

**LMI/OGN/OPS/MECH/012/**

**SEPTEMBER-2024**

**Approved for Implementation by HOP (SIMHADRI)**

## LMI APPROVAL DOCUMENT

|   |                     |                                                                      |
|---|---------------------|----------------------------------------------------------------------|
| 1 | LMI NAME            | Deaerator Storage Vessels - Reinspection and Monitoring Requirements |
| 2 | LMI No.             | LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-07, Sep. 2024                     |
| 3 | Reference Document  | COS-ISO-00-OGN/OPS/MECH/012 Rev. No.: 3, Aug. 2024                   |
| 4 | Superseded Document | LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-06, Jul. 2022                     |
| 5 | Prepared By         | Name: Sahil Sankalp,<br>Manager (TMD)                                |
| 6 | Checked By          | Name: Nilanjan Mandal,<br>AGM (TMD)                                  |
| 7 | Reviewed By         | Name: Prasanta Kumar Jena<br>G M (O&M)                               |
| 8 | Approved By         | Name: S. K. Sinha<br>H O P (Simhadri)                                |



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|         |                                                                      |
|---------|----------------------------------------------------------------------|
| TITLE   | Deaerator Storage Vessels - Reinspection and Monitoring Requirements |
| LMI No. | LMI/OGN/OPS/MECH/012/                                                |

## 1.0 INTRODUCTION

Following failures of Deaerator Storage Vessels overseas, utilities have commenced reinspection to identify any significant defects during a vessels service life. Most of the defects found were manufacturing faults usually associated with welds. During running also passing of spray nozzles, damages in perforated sheets / trays, erosion in drip connection area etc. were noticed while unit in service. Deaerator storage vessels should be monitored by periodic reinspection to ensure that cracking does not develop after long service. There should also be reinspection of vessels where defects of a more serious nature have been found and the vessel left in service, or found and the defect ground out, or repaired. This reinspection is in addition to the normal visual inspections carried out at each statutory inspection.

## 2.0 SUPERSEDED DOCUMENTS

LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-06, Jul. 2022

## 3.0 DETAILS OF REVISION

Following changes have been incorporated in the current revision –

| SI No | LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-06, Jul. 2022 | LMI/OGN/OPS/MECH/012/ISSUE-1, REV.-07, Sep. 2024                      |
|-------|--------------------------------------------------|-----------------------------------------------------------------------|
| 1.    | -                                                | New Clause 4.1.1 (h): Water spillage through vent have been included. |
| 2.    | -                                                | New Clause 4.1.1 (i): Civil foundation checking has been included.    |
| 3.    | -                                                | New Clause 4.1.1 (j): Standpipe inspection have been included.        |
| 4.    | -                                                | Appendix-2: Checklist has been included.                              |

## 4.0 PROCESS:

### 4.1 Reinspection:

4.1.1 This guidance note illustrates the need for detailed investigation as detailed below. First such inspection shall be made during first overhaul of Turbine to establish datum and type of defects in various vessels. **The inspection of Deaerator storage vessel shall be taken up during every unit overhaul once in two years.** The details of inspection to be carried out during every unit overhauling is given bellow:

| TITLE   | Deaerator Storage Vessels - Reinspection and Monitoring Requirements |
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| LMI No. | LMI/OGN/OPS/MECH/012/                                                |

- a) Visual inspection of Internal surface of deaerator for pitting, crack & deposit etc. Manual cleaning of Vessel internals.
- b) Visual inspection of weld joints of longitudinal and transverse of deaerator shell after thorough cleaning. 1/3rd of main seam weld joint may be covered during each overhaul progressively and to maintain a data base of record so that all weld joints may be covered in three inspection cycles. Repairing of weld defects as per OEM Guidelines.
- c) Inspection of SS Perforated Sheets/ Nozzle plate / trays for any damages, crack, looseness /falling, cleanliness. Carry out repair / replacement of damaged parts.
- d) Inspection of spray nozzles for any damaged, looseness /falling, Cleanliness. Repair / replacement of damaged spray nozzles.
- e) Inspection of supporting structure of steam heating line header for any crack, weld defects, looseness. Carry out necessary repair works.
- f) Inspection of all drip line connection perforated pipes/ nozzle connection for any damage/ cracks. Inspection of Deaerator wall thickness for any erosion and thickness mapping of deaerator vessel near drip connections. Carry out repair & replacement of damaged parts
- g) Inspection of Deaerator continuous vent orifices, Deaerator vents line root welding joint. Carry out DPT & rectification of defects.
- h) Prior to unit outage, check for any water spillage from deaerator vent line. If observed, then the reasons like increase in orifice size, controlled vent v/v opening, baffle plate inside D/A etc. for the same to be identified & rectified.
- i) Carry out inspection of civil foundation and foundation bolt. Ensure availability of provisions for free expansion of FST on both sides.
- j) Inspection of standpipes weld joints, servicing of stand pipe isolating valves and vent valves to be carried out. Availability/Healthiness of level gauges to be ensured.

The detailed inspection carried out shall be recorded as given in Appendix 01. Check list as per Appendix 02 to be ensured.

Where large numbers of similar defects are present, it is sufficient to inspect a representative sample.

4.1.2 The technique to be used for re inspection should be of NDT (DPT or Magnetic particle inspection).

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4.1.3 As the areas to be inspected will have been ground previously it is anticipated that the required surface finish of 3.2 um Ra will be achieved by means of a flywheel or soft abrasive disc. Wherever possible the use of a hard grinding disc should be avoided as this may result in the removal of existing defects and reveal previously embedded defects making comparisons invalid.

4.1.4 In addition to the reinspection of known defects or repairs, a visual inspection should be made of the inner surface within 150 mm of the main welds. Evidence of the remnants of any tack welds associated with temporary attachments should be noted.

If any such welds are found, they should be ground flush with the inner surface. Remedial measures should be consistent with those applied during the datum inspections.

If datum inspection has not been done at site, the same shall be carried out and the results compared in subsequent inspections.

#### **4.2 Monitoring:**

4.2.1 For newer units, the Deaerator storage vessels of 1st unit shall be inspected during first mandatory inspection. If major defects are noticed, then all the units under erection /commissioning shall be inspected.

4.2.2 The inspection should be applied progressively to 1/3rd main seam joint of the vessel during each overhaul and to be kept as record. A small sample of areas containing stress raisers such as penetrations and internal furniture attachment welds should also be inspected if possible. Wherever possible, the areas inspected should be those examined during the datum inspection.

As most of the areas to be inspected will have been ground previously, it is anticipated that the required surface finish of 3.2 um Ra will be achieved by means of soft abrasives. Use of hard grinding discs should be avoided since this may remove small defects, which have initiated recently.

#### **4.3 Identification of defects:**

A sample of the defects left in service should be pop marked to show their location and extent. The defects and pop marks should then be sketched or photographed. The depth of any grinding or regrinding should be accurately recorded. Repair to be taken up as per procedure recommended by OEM

#### **4.4 Implementation:**

4.4.1 Inspections of the vessels should be carried out at each 'thorough examination' except where running hours and start are significantly less than 15,000 and 300

|                |                                                                             |
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respectively. For these low service cases, exemption may be allowed provided that running hours and starts are not expected to exceed more than 25,000 and 500 respectively by the next inspection opportunity.

4.4.2 The defects noticed during the above inspection shall be repaired as per OEM guidelines. A Joint protocol shall be made by Turbine maintenance along with FQA for all weld repairs & NDT inspection of weld joints.

## 5.0 RECORDS TO BE MAINTAINED

| Sl No. | Record Description                                                                                                                                                                                                                                                                    | Maintained By                                                                                                                                                                                                                                                                                                                           | Frequency           | Remarks                                                                         |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------|
| 1      | Records must be kept for all inspections and should contain the following data: <ul style="list-style-type: none"> <li>• Date of inspection</li> <li>• Details of inspection carried out</li> <li>• Exception of the inspection.</li> <li>• Details of repair carried out.</li> </ul> | Responsibility of carrying out above inspection shall be with TMD Department. Joint protocol shall be made by TMD along with FQA for all weld repairs & NDT inspections.<br><br>Joint protocols shall be made with Operation and Chemistry department after attending all defects & thorough cleaning before final box up of Deaerator. | As mentioned Above. | All defects noticed during above inspection shall be rectified during overhaul. |

## 6.0 REVIEW

This LMI document shall be reviewed by the HOD-TMD on a 2 yearly basis or as necessary and the same shall be approved by HOP.

## 7.0 DISTRIBUTION OF LMI

Soft copy of the LMI shall be uploaded in the EEMG dept. website on Simhadri intranet LAN by the EEMG Department.

|                |                                                                             |
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## 8.0 CHECKLIST

### Checklist 01 - DEAERATOR INSPECTION PROTOCOL

#### A. Internal surface of deaerator

- Pitting
- Cracks
- Deposits

#### B. Preheat steam pipe welding

#### C. SS Perforated Sheets:

- Damages
- Looseness / Falling
- Cleanliness
- Holding Bolts

#### D. Nozzles:

- Damages
- Looseness / Falling
- Cleanliness

#### E. Drip line connection perforated pipes/ nozzle connection.

- Damages
- Crack
- Erosion

#### F. Any Other Observation

- Deaerator Continuous vent
- Any other points

|                |                                                                             |
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## Checklist 02 - DEAERATOR INSPECTION CHECKLIST

| Date of Last Inspection:                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Date of Current Inspection: |         |
|---------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|---------|
| Inspection to be done every 2 years, so Next Inspection Date: |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                             |         |
| S. No                                                         | Points to be observed                                                                                                                                                                                                                                                                                                                                                                                                                              | Checked (Yes/No)            | Remarks |
| 1.                                                            | Check for any reported increase in feed water Dissolved Oxygen prior to Overhaul                                                                                                                                                                                                                                                                                                                                                                   |                             |         |
| 2.                                                            | Prior to unit outage, check for any water spillage from vent lines                                                                                                                                                                                                                                                                                                                                                                                 |                             |         |
| 3.                                                            | Carry out visual inspection of Deaerator internal for visible defects such as pitting, cracks, etc.                                                                                                                                                                                                                                                                                                                                                |                             |         |
| 4.                                                            | Carry out inspection and NDT of main seam Weld joints and maintain record (1/3rd of all joints to be covered on 1 occasion)                                                                                                                                                                                                                                                                                                                        |                             |         |
| 5.                                                            | Check following connections for cracks, weld defects, erosion, etc.: <ul style="list-style-type: none"><li>• Heating line header &amp; support</li><li>• Drip Line connections, perforated pipe, support, etc.</li><li>• BFP Recirculation line connections &amp; support</li><li>• SS Perforated Sheets/ Nozzle plate / trays for any damages, crack, looseness /falling, cleanliness. Rectification work to be done wherever required.</li></ul> |                             |         |
| 6.                                                            | Check condition of spray nozzles, tightness of mounting bolts, healthiness of gaskets, and replenish fallen nozzles                                                                                                                                                                                                                                                                                                                                |                             |         |
| 7.                                                            | Check condition of Distribution trays for damages and carry rectification work, if required                                                                                                                                                                                                                                                                                                                                                        |                             |         |
| 8.                                                            | In case of visible erosion at D/A internal wall, carry out thickness survey in proximity of affected area to determine wall thickness. Inspection of Deaerator wall thickness for any                                                                                                                                                                                                                                                              |                             |         |

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|     |                                                                                                                                                                          |  |  |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
|     | erosion and thickness mapping of deaerator vessel near drip connections                                                                                                  |  |  |
| 9.  | Check tightness of all mounting bolts, clamps, fittings, fasteners, etc. that are located inside D/A                                                                     |  |  |
| 10. | Carry out DPT of any repair welding carried out inside D/A before box-up                                                                                                 |  |  |
| 11. | Check condition of Manhole and all flange gaskets. Replace if required                                                                                                   |  |  |
| 12. | Carry out inspection of civil foundation and foundation bolt. Ensure availability of provisions for free expansion of FST in both side                                   |  |  |
| 13. | Inspection of standpipes weld joints, servicing of standpipe isolating valves and vent valves to be carried out. Availability/Healthiness of level gauges to be ensured. |  |  |

\*Last Inspection date to be mentioned suitably on equipment



|                |                                                                             |
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## **9.0 ANNEXURE**

Annexure 1 - DP Test of Feed Storage Tank

|                |                                                                             |
|----------------|-----------------------------------------------------------------------------|
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**Annexure 1****DP TEST OF FEED STORAGE TANK**

Visual Inspection and Dye penetrate test (DPT) of followings were carried out.

Feed storage tank DPT carried out on site welded & factory welded circumferential joints & factory welded longitudinal joints as per annexure-A and \_\_\_\_\_ (no/few) cracks were observed and found \_\_\_\_\_ (satisfactory/not satisfactory).

DP KIT USED:

PARTICULARS OF PENETRANT : ORION 115P, Mfg. date: \_\_\_\_\_ / DOE: \_\_\_\_\_

PARTICULARS OF DEVELOPER : ORION 115P, Mfg. date: \_\_\_\_\_ / DOE: \_\_\_\_\_

PARTICULARS OF REMOVER : ORION 115P, Mfg. date: \_\_\_\_\_ / DOE: \_\_\_\_\_

DP test of FST checked on:

TESTING AND ACCEPTANCE STANDARD : ASTM-E-165

DWELL TIME : 15 MIN

OBSERVATION / DEFECT NOTICED : \_\_\_\_\_ (No/few) Cracks observed

TYPE OF DEFECTS : \_\_\_\_\_ (NIL/Type of defect)

LOCATION OF DEFECTS : \_\_\_\_\_ (NIL/location of defect)

RESULTS : \_\_\_\_\_ (Satisfactory/not satisfactory)

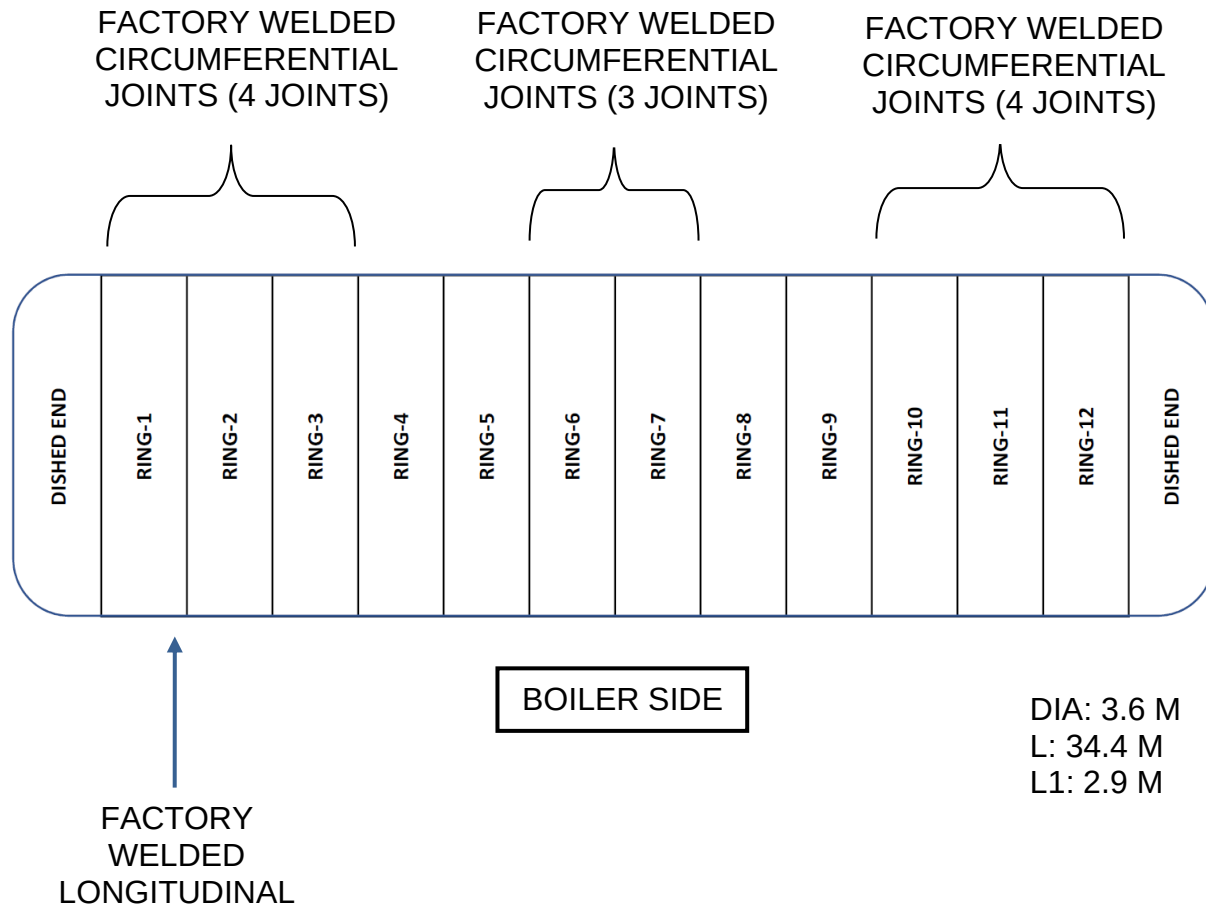
ACCEPT / NOT ACCEPT :

|                         |          |          |
|-------------------------|----------|----------|
| CHECKED & CERTIFIED BY: |          |          |
| SIGNATURE               |          |          |
| NAME                    |          |          |
| DESIGNATION             |          |          |
| DEPT                    | NTPC-TMD | NTPC-FQA |

|                |                                                                             |
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**DEAERATOR – FEED STORAGE TANK CROSS SECTION**

**SWYD SIDE**



NOTE: ABOVE JOINTS WHICH AS UNDERGONE DPT TO BE MARKED ON THE ABOVE DRAWING.